TCS — Aptitude Test 3

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Question 51. The number of times a bucket of capacity 4 liters to be used to fill up a tank is less than the number of times another bucket of capacity 3 liters used for the same purpose by 4. What is the capacity of the tank?

of work = 9/3 = 3 days.
Work done by George in 3 days = 1/5
Work done by George in 1 day = 1/15
Remaining work = 1 – (3/5) = 2/5
Time taken by both George and Paul together to complete 2/5th of the work = 4 days
Time taken for them to complete full work = 4 x 5/2 = 10 days.
Work done by George and Paul together in one day = 1/10

Solution:
LCM of 4 and 3 is 12.
For a 12 liter tank, we need 3 buckets of four liter and 4 buckets of 3 liter.
The number of 4 liters bucket is 1 less than the number of 3 liter buckets required.
For the number of 4 liters buckets to be less than the number of 3 liter buckets by 4, the tank should be 4 times of the 12 liters tank.
The capacity of the required tank is = 4 x 12 = 48 liters.
For 48 liters, 4 liter buckets required is 12 and 3 liter buckets required is 16.

1
G
=
1
10

1

+

P

T C S

1

1

15
=
1

10

+

P

P = 30 days.

Question 52. A cistern consists of 3 pipes. By the help of 1st and 2nd pie the tank gets filled in 10 hours and 15 hours respectively but by the help of 3rd pipe tank becomes empty in 5 hours. At 8 a.m., 1st pipe was opened. At 9 a.m., 2nd pipe was also opened with 1st. At 10 a.m., 3rd pipe was also opened with 1st and 2nd. When the tank becomes filled or empty?

Question 53. Arun and Vinay together can do a piece of work in 7 days. If Arun does twice as much work as Vinay in a given time how long will Arun take to do work?

Solution:
Work done by Arun and Vinay together in one day = 1/7.
Let us assume that the work done by Vinay in one day = 1/V
So, work done by Arun in one day = 2/V
So,

1
V
=
1
7

2

V
+

V = 21 days.
Arun will complete the work in half time of Vinay.
Time taken by Arun = 21/2 = 10.5 days.

Question 54. Brilliant software company, Chennai has been doing an excellent business in the last four years. The company went on a recruitment spree from among the engineering colleges in and around Chennai. They recruited people from ECE, CSE, IT streams. All programmers are of equal respect. They receive equal salaries and perform equal load of work. Suppose 15 such programmers take 15 minutes to write 15 lines of code in total. How long will it take for 84 programmers to write 84 lines of code in total?

Solution:
Work done is directly proportional to resource and time taken.
Here the work done (W) is number of lines of code.
Resources (R) is the number of programmers.
Time (T) is the number of minutes.

W1
W2

=
R1T1
R2T2
15
84
=
15 x 15
84 x T2

■

T2 = 15 minutes.

Question 55. There are 4 machines namely P, Q, R and S in a factory. P and Q running together can finish an order in 10 days. If R works twice as P and S works 1/3 as much as Q then the same order of work can be finished in 6 days. Find the time taken by P alone to complete the same order.

Solution:

1
P
=
1
10

1

Q
+
---(1)

1
R
=
1
6

1

S
+

T C S

2
P
=
1
6

1
3Q
+
---(2)

■

Assume that (1/P) = x and (1/Q) = y
From equ. (1), we get
x + y = 1/10 ---(3)
From equ. (2), we get
2x + (y/3) = 1/6 ■6x + y = ½ ---(4)
By solving (3) and (4), we get
y = 1/50 and x = 2/25
x = (1/P) = (2/25)
P = 25/2 = 12.5 days.

Question 56. If two-third of a bucket is filled in one minute then the time taken to fill the bucket completely will be?

Solution:
2/3rd of a bucket is filled in 1 minute.
Time taken to fill 1/3rd of the bucket
= ½ minutes = 30 seconds.
Time taken to fill the full bucket
= 30 x 3 = 90 seconds = 1.5 minutes.

Question 57. If 10 lions can kill 10 deer in 10 minutes how long will it take 100 lions to kill 100 deer.

Solution:

W1
W2

=
R1T1
R2T2
10
100
=
10 x 10
100 x T2

■

T2 = 10 minutes.

starts at 7pm = 2 x 40 = 80km
Time taken for car 2 to meet car 1
=80/(60 – 40) = 80/20 = 4 hours.
Time taken for car 2 to be 9km ahead of car 1 = 9/(60/40)
= 9/20
= 0.45 hours.
= 0.45 x 60 = 27 minutes.
Total time taken for car 2 to be ahead of car 1 = 4 hours 27 minutes.

Question 58. 33

58. A cistern can be filled by one of the pipes in 30 min and by the other in 36 min .both pipes are opened for a certain time but being partially clogged only 5/6 of the quantity of water flows though the former and only 9/10 through the later the obstruction however being suddenly removed the cistern is filled in 3.5 minutes form that moment. How long was it before the full of water began?

Question 59. Car A leaves city C at 5pm and is driven at a speed of 40kmph. 2 hours later another car B leaves city C and is driven in the same direction as car A. In how much time will car B be 9 km ahead of car A if the speed of car is 60kmph?

Solution:
Distance travelled by car 1 before car 2

= 2000/S2
From the question
1000/S1 = 2000/S2 – 6 ---(1)
By solving (1) and (2), we get
S1 = 250 and S2 = 200
Time taken for Ram to complete the race
= 2000/250 = 8 minutes
Time taken for Shakil to complete race
= 2000/200 = 10 minutes.

Question 60. Jake is faster than Paul. Each walk 24 km. The sum of their speeds is 7kmph and the sum of the times taken by them is 14 hr. Then Jake speed is equal to:

Solution:
Let N be the number of minutes for which the clogs were there in the two pipes.
Time taken to fill the tank by pipe 1 with the clog = 30 x (6/5) = 36 minutes.
The fraction of tank filled by pipe 1 in one minute, when the clog is there is,
= 1/36
Total fraction of tank filled by pipe 1 is:

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Solution:
Let the speed of Jake and Paul be x and y respectively.
Given, x + y = 7
---(1)
Time taken by Jake = 24/x
Time taken by Paul = 24/y
(24/x) + (24/y) = 14
(24x + 24y)/xy = 14
xy = 24(x+y)/14 = 168/14
xy = 12
---(2)
By solving (1) and (2)
We get, x = 4 and y = 3
Speed of Jake = 4 kmph.

N

Question 61. Ram and Shakil run a race of 2000 meters. First, Ram gives Shakil a start of 200 meters and beats him by one minute. If Ram gives Shakil a start of 6 minutes Ram is beaten by 1000 meters. Find the time in minutes in which Ram and Shakil finish the race separately.

N

Question 62. If ABC is a quarter circle and a smaller circle is inscribed in it; if radius of quarter circle is 1.414units. Find the radius of smaller circle.

Solution:
The following diagram represents the given question.

RE is perpendicular to BC.
REDC form a square.
From the diagram, RE = EC = r
RC = r√2. (Diagonal of a square = a√2)
r + r √2 = √2
r(1 + √2) = √2
r = √2/(1+ √2)
r = 0.586 units.

Question 63. 2 identical circles intersect so that their centers and the points at which they intersect form a square of side 1cm. The area in sq. cm of the portion that is common to 2 circles is:

Solution:
The following diagram represents the question.

a
b

c
d

Area of the square = 1 x 1 = 1sq.cm
Area of half of the square = 0.5 sq.cm
Area of the section formed by 'adc' in the circle = (1/4) x πr2 = (¼) π
Area of the shaded part below the diagonal of the square 'abcd'
= π/4 – 0.5
= 0.285 sq.cm
Similarly area of the shaded part above the diagonal = 0.285 sq.cm
Total shaded area = 2 x 0.285
= 0.57 sq.cm.

Question 64. Usha bought a linen cloth and rope to build a tent. If the rope is 153 m long and it is to be cut into pieces of 1m length, then how many cuts are to be made to cut the ropes into 153 pieces?

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Solution:
In a two meter rope, if one cut is made, there will be two pieces.
In a three meter rope, if two

cuts are
made, there will be three pieces.
Number of cuts required =
Number of pieces – 1.
For 152 pieces, the number of cuts
= 153 – 1 = 152.

Question 65. Given 3 lines in the plane such that the points of intersection form a triangle with sides of length 19, 19 and 19, the number of points equidistant from all the 3 lines is?

Solution:
For an equilateral triangle, the point
which is equidistance from all the lines
is the in-center of the triangle.
In-center is the exact middle of the
equilateral triangle.
Number of points = 1.

Question 66. In the town of UnevenVille, it is a tradition to have the size of the front wheels of every cart different from that of the rear wheels. They also have special units to measure cart wheels which is called uneve. The circumference of the front wheel of a

Question 67. Rectangular tile each of size 70cm by 30cm must be laid horizontally on a rectangular floor of size 110cm by 130cm,such that the tiles do not overlap and they are placed with edges touching against each other on all edges. A tile can be placed in any orientation so long as its edges are parallel to the edges of floor. No tile should overshoot any edge of the floor. The maximum number of tiles that can be accommodated on the floor is:

Question 68. If length, breadth and height of cube are decreased, decreased & increased by5%,5%,20% respectively then what will be the impact of surface area of cube?

cart is 133 uneves and that of the back wheel is 190 uneves. What is the distance traveled by the cart in uneves, when the front wheel has done nine more revolutions than the rear wheel?

Solution:
Surface area of cube = 6a2
Assume that the side of a cube = 100.
Surface area of cube = 60,000
The cube will become a cuboid after changing the dimensions as per the question.
Sides of the cuboid:
Length, l
= 95
Breadth, b
= 95
Height, h
= 120
Surface area of cuboid = 2(lb + bh + hl)
= 2(95x95 + 95x120 + 120x95)
= 63,650
Percentage change = (3650/60000)100
= 6.08% increase.

Solution:
The difference b/n the circumference of the two wheels = 57
For every one revolution of the back wheel, the front wheel has to make one revolution + 57 uneves.
The extra uneves covered by the front wheel with 9 more revolutions = 9x133
= 1197 uneves
The number of 57 extra uneves in 1197
= 1197/57 = 21.
So, the back wheel has made 21 revolutions. The total distance covered
= 21 x 190 = 3990 uneves.

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Question 69. On planet korba, a solar blast has melted the ice caps on its equator. 9 years after the ice melts, tiny planetoids called echina start growing on the rocks. Echina grows in the form of circle, and the relationship between the diameter of this circle and the age of echina is given by the formula d = 4*(t-9) for t>9, where d represents the diameter in mm and t the number of years since the solar blast. Jagan recorded the radius of some echina at a particular spot as 7mm. How many years back did the solar blast occur?

Solution:
The following diagram represents the above description.

130

Solution:
Radius = 7mm; Diameter = 14mm
So, 14 = 4(t – 9)
t – 9 = 14/4 = 3.5
t = 12.5 years.

110

Question 70. A boy wants to make cuboids of dimension 5m, 6m and 7m from small cubes of .03 m3. Later he realized he can make same cuboids by making it hollow. Then it takes some cubes less. What is the number of the cubes to be removed?

The maximum number of tiles that can be placed =5(without breaking the tiles)
With breaking= 110x130/30x70 = 6

Solution:
The volume of the whole cuboid
= 5 x 6 x 7 = 210 m3

After removing the outer shell, the cuboid which has to be originally removed will be inside.
The volume of that cuboid is
(5 – 0.03)x(6 – 0.03)x(7 – 0.03)
= 4.97 x 5.97 x 6.97

Volume = 206.8 m3

This 206.8 is the volume that has to be removed from 210m3 cube.
The number of small cubes in 206.8m3

cube is
=206.8/0.03 = 6893 cubes.

Question 71. A person is writing all the 4 digit numbers. How many times he will use 2?

Solution:
From 00 to 99 there are 20 2's.
There are 90 sets of 00 to 99 as the last two digits from 1000 to 9999.
Number of 2's in the last two digits of all 4 digit numbers = 90 x 20 = 1800
From 000 to 999, there are 100 2's in the hundreds place.
There are 9 sets of 000 to 999 from 1000 to 9999.
The number of 2's in the hundreds place = 9 x 200 = 1800
From 1000 to 9999, there are 1000 numbers in which the thousands place is 2.
Total number of times, two is written = 1800 + 1800 + 2000 = 5600

Question 72. How many 6 digit even numbers can be formed from digits 1, 2, 3, 4, 5, 6, and 7 so that the digit should not repeat and the second last digit is even?

Solution:
There are three even numbers in the given set of numbers (2, 3 and 6).
The number of ways in which we can form a pair of even numbers for the last two digits out of three even numbers is:
3P2 = 3 x 2 = 6. They are
24, 26, 42, 46, 62 and 64

There are six cases and in each six case there are 5 numbers remaining, which can be arranged in 5! ways. (Without repetition).
The total number of arrangements = 6 x 5! = 720.

Question 73. The letters in the word "PLACES" are permuted in all possible

ways
and
arranged in the alphabetical order. Find the word at 48 position.

Solution:
Arrange the given letters in alphabetical order as below.
ACELPS.
Assign numbers as given below.
A = 1; C = 2; E = 3; L = 4; P = 5; S = 6
When we arrange the numbers in ascending order, the first number will be 123456.
We have to find the 48th term.
48 is a multiple of 4!.
24th term will be the reversal of the last 4 digits.
24th term = 126543
25th term = 132456
48th term will be the reversal of last 4 digits of 25th term.
48th term = 136542
Arrange the letters according to the numbers above.
48th word will be
AESPLC.

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Question 74. What is the sum of all 5 digit numbers which can be formed with the digits 0,1,2,3,4 without repetition?

Solution:
Total number of five digit numbers that can be formed:
First digit can be filled in 4 way.(except zero).
Second digit can be filed in 4 ways (including zero).
Third digit can be filled in 3 ways.
Fourth digit in 2 ways and
Fifth digit in 1 way.
Total ways = 4x4x3x2x1 = 96 ways.

There are no letters repeated in the given word.
Combine A, U and E as one element.
The number of elements available now = 6.
Number of ways of arranging these 6

letters = 6!.
Number of ways of arranging the 3 vowels internally = 3!
Total ways of arranging = 3! X 6!
= 6 x 720 = 4320

Among the 96 five digit numbers,
24 numbers will start with 1,
24 numbers will start with 2,
24 numbers will start with 3,
24 numbers will start with 4,
For numbers starting with 1, the unit place will be either 0 or 2 or 3 or 4.
Out of these 24 numbers starting with 1, we will have
6 numbers ending with 0,
6 numbers ending with 2,
6 numbers ending with 3, and
6 numbers ending with 4.
Sum of these unit place numbers
= 6(0+2+3+4) = 54.
Sum of tens digits = 540
Sum of hundreds digits = 5400
Sum of thousands digits = 54000
Sum of ten thousands digits
= 10000 x 24 = 240000
Sum of the 24 numbers starting with 1
= 54+540+5400+54000+240000
= 299994
Similarly, sum of units digits of numbers starting with 2 = 6(0+1+3+4) = 48
Sum of all numbers starting with 2
= 48 + 480 + 4800 + 48000 + 480000
= 533328
Similarly sum of unit digits of numbers starting with 3 = 6(0+1+2+4) = 42
Sum of all numbers starting with 3
= 42 + 420 + 4200 + 42000 + 720000
= 766662
Similarly, sum of unit digits of numbers starting with 4 = 6(0+1+2+3) = 36
Sum of all numbers starting with 4
= 36 + 360 + 3600 + 36000 + 960000
= 999996
Sum of all the 96 numbers
= 299994+533328+766662+999996
= 2599980

Question 75. How many words can be formed using the letters of the word

“DAUGHTER” so that all the vowels occur together?

Solution:
The vowels in the letter DAUGHTER are A, U and E.

For each case the remaining 7 digits out of 9 digits can be arranged in 57 ways, since thee repetition is allowed.
Total number of arrangements
= 57 x 5 = 58

= 390625 ways.